

# **GaN/Si Summit: process reliability status**

Peter Ersland  
18 October 2016

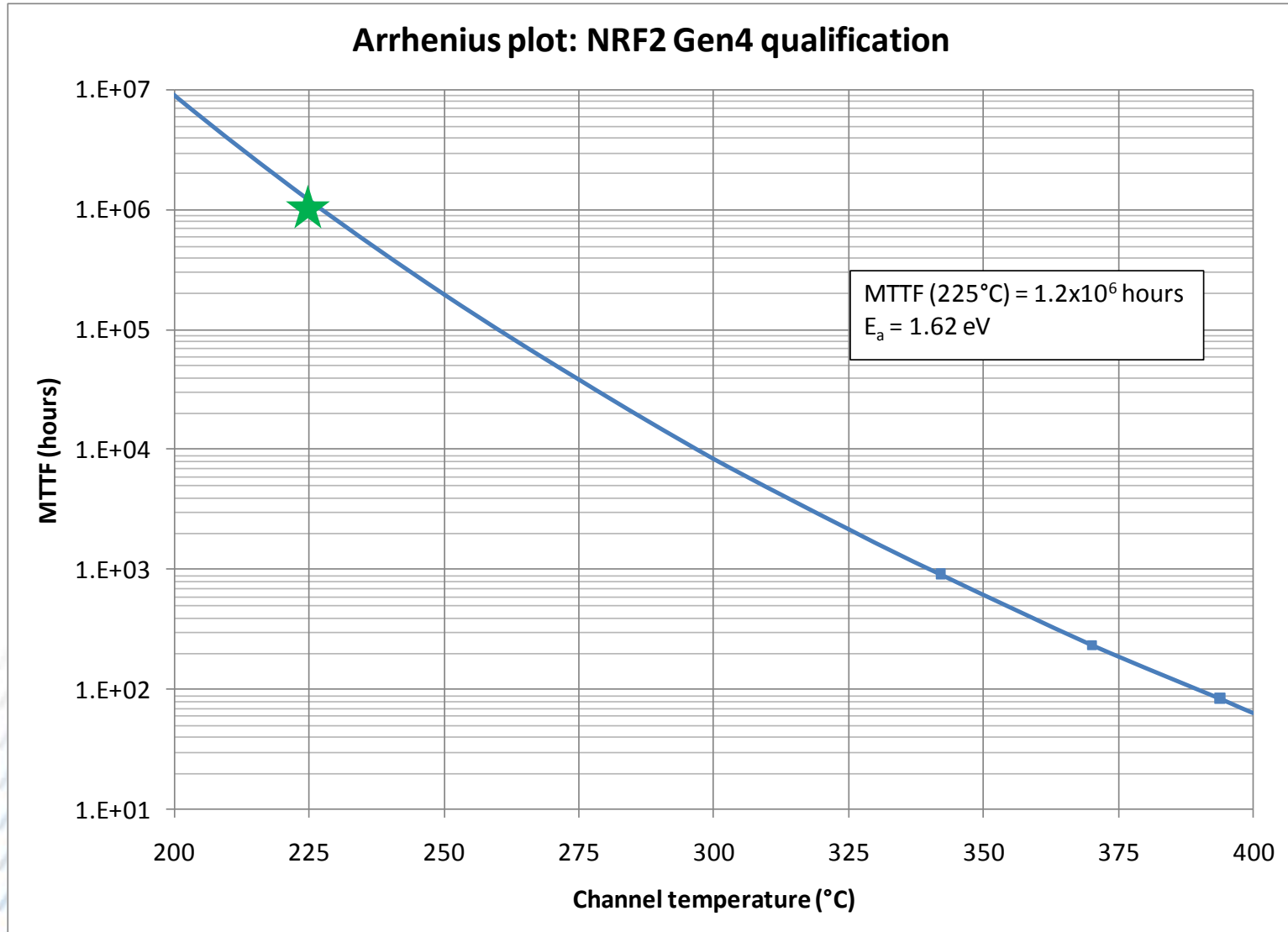


# **GaN/Si process reliability status**

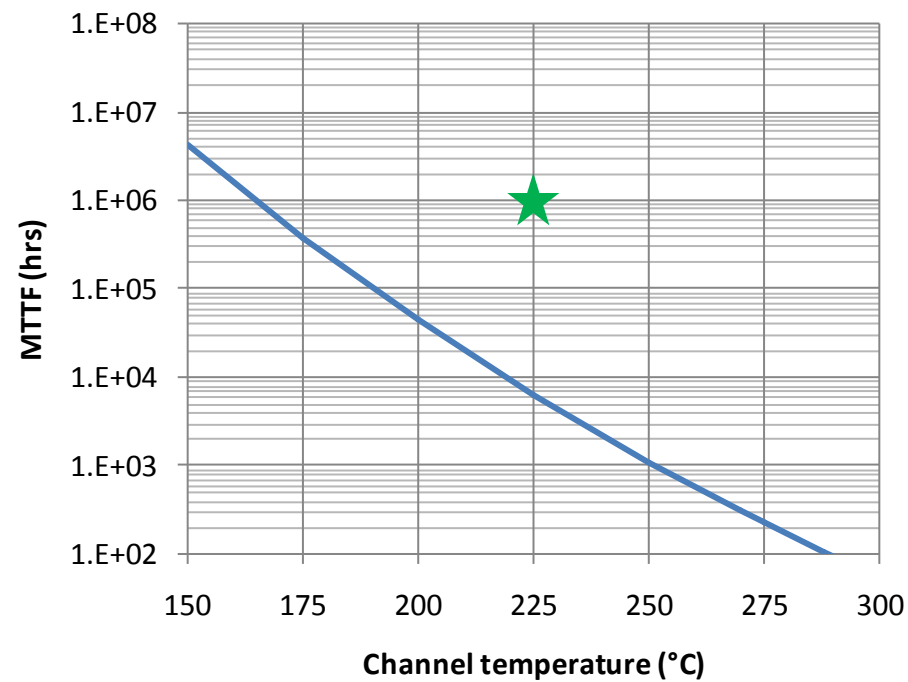
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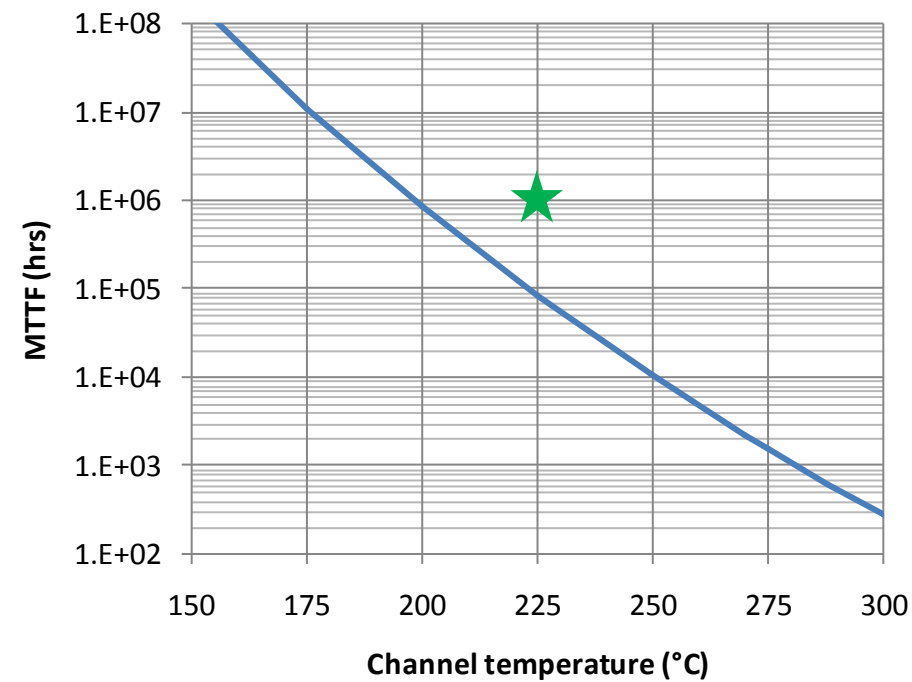
- IQE epi material (100  $\mu\text{m}$  wafer diameter) – PASSED
  - Tested using GCS 0.5  $\mu\text{m}$  process; 0.4 mm device size
- WIN NP45-20 – FAILED
  - Rapid drain current degradation
  - MTTF more than 10x below requirement
- GCS 0.4  $\mu\text{m}$  process – Considered qualified
  - Identical to 0.5  $\mu\text{m}$  process used for IQE epi qual and NTX qual (2013)
  - Additional tests currently in process: 0.4  $\mu\text{m}$  Lg, 2.04 mm device size
  - Wafer-to-wafer uniformity (wrt reliability) is a concern
- MTS GSI40 – Reliability tests in process
  - Very positive results to date
  - 0.6 mm periphery device; will test larger devices when available
- STM – Minimal testing performed to date
  - Only 6 parts; significant early reduction in  $I_{\text{max}}$



NP45-20: Rel lot 1



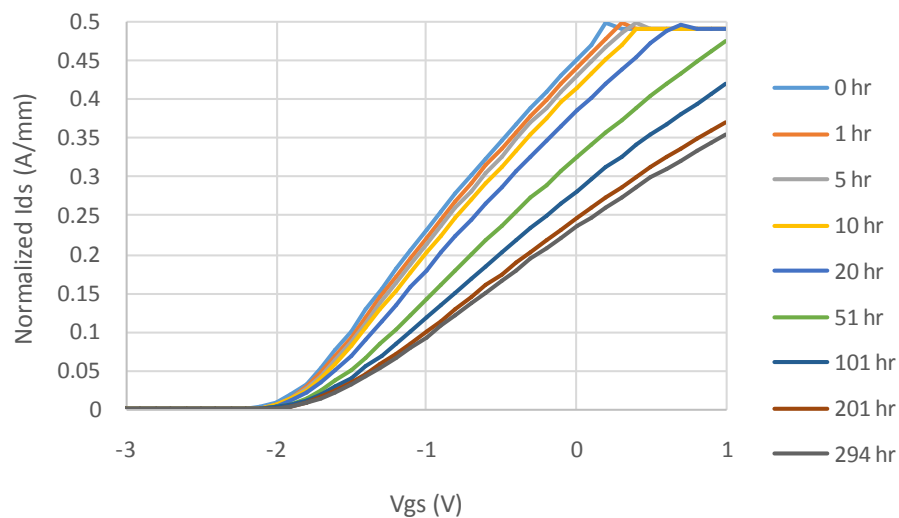
NP45-20: Rel lot 2



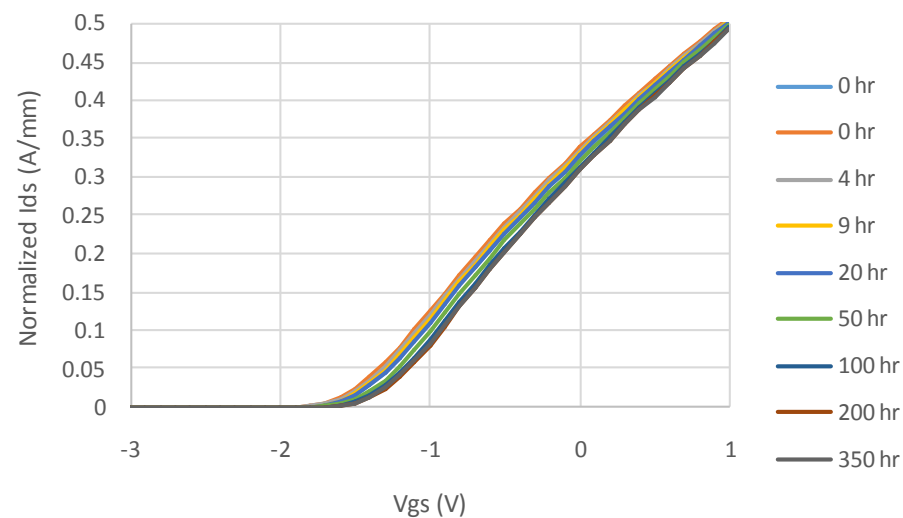
# Transfer curves: WIN vs Lowell



XFER 606-423



XFER 2081-605



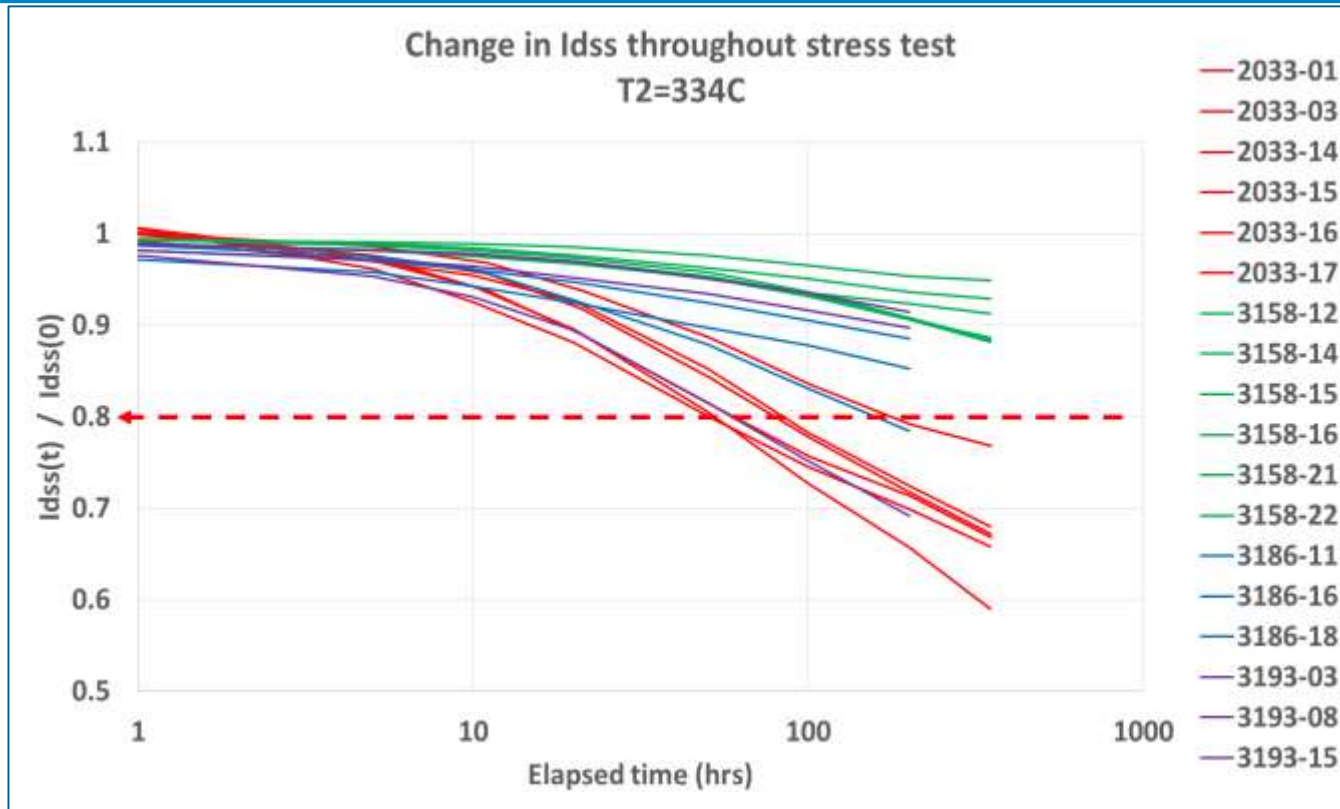
WIN NP45-20

T<sub>ch</sub> = 271°C

Lowell GSI40

T<sub>ch</sub> = 327°C

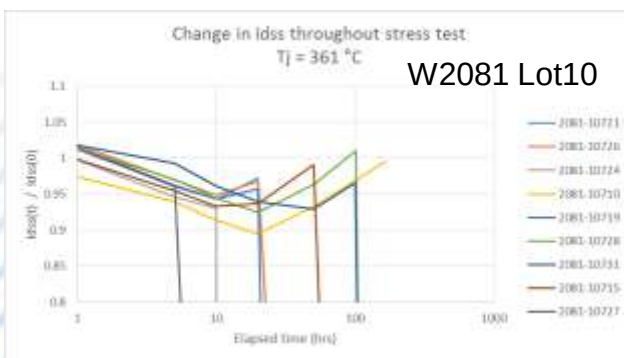
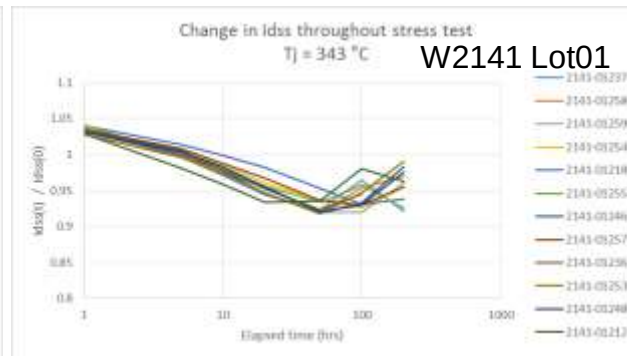
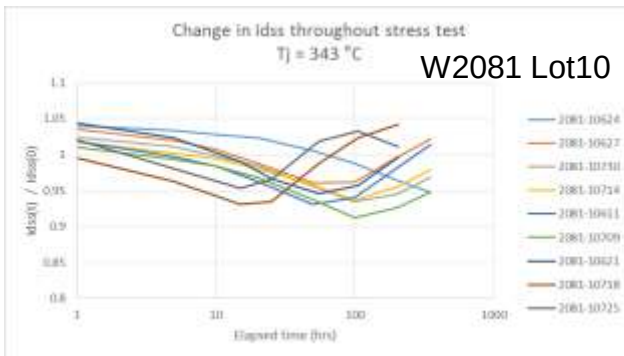
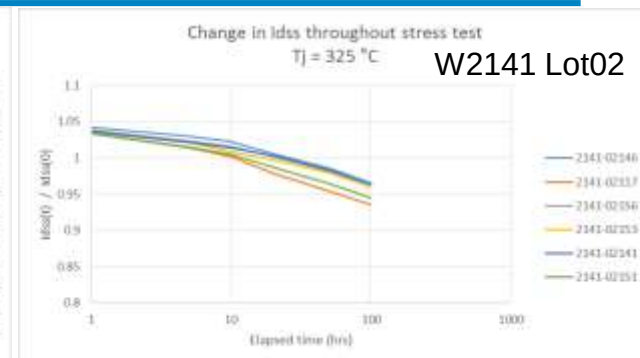
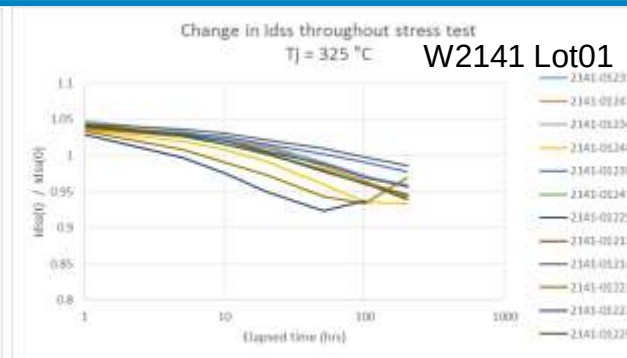
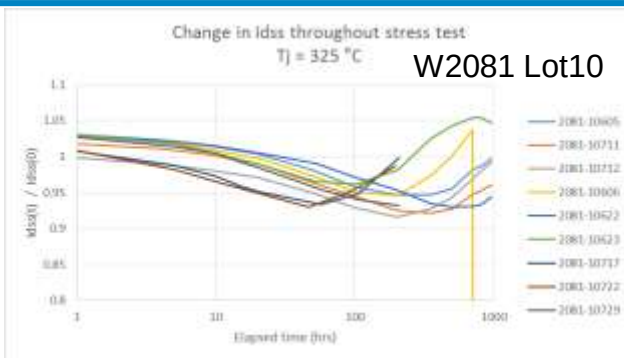
# GCS reliability tests: uniformity concerns



- Devices from 4 different wafers (2 wafer lots)
- Arrhenius curve from IQE epi qual suggests  $\text{MTTF} > 1,000 \text{ hrs}$  at  $334^{\circ}\text{C}$ 
  - Only one of these wafers clearly meets this target
  - Note difference in device sizes: 2.04 mm here vs 0.4 mm in IQE epi qual



# Lowell process – very stable

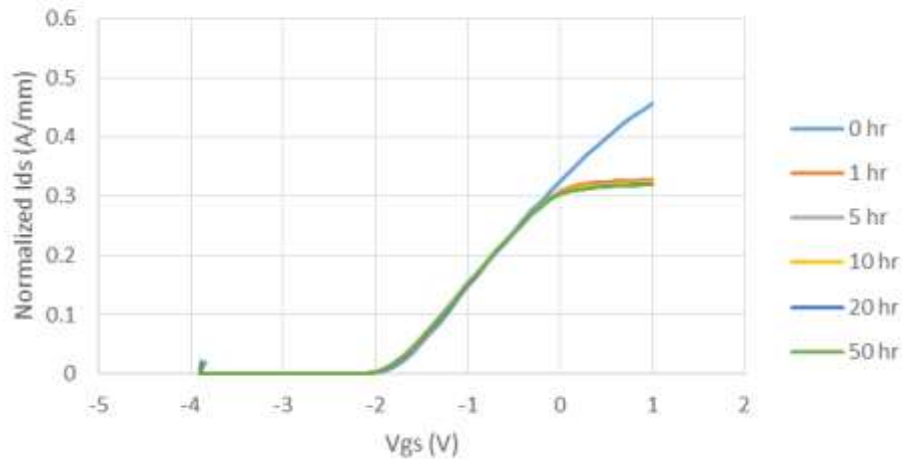


- Tests on 2 finger, 0.6 mm device
- Very high channel temperatures; may be introducing new failure mechanisms
  - Exhibiting better drain current reliability than either WIN or GCS – both stability and uniformity



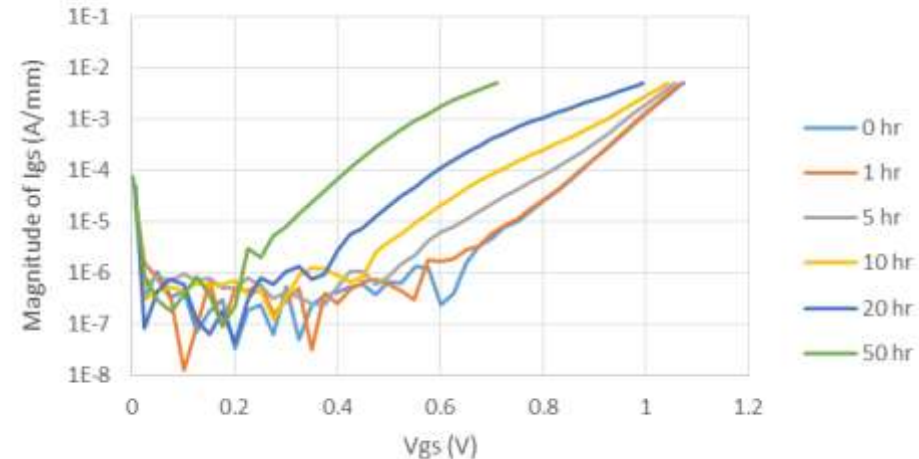
XFER STM-A2

Tch = 305°C



VGSP STM-A2

Tch = 305°C



- Rapid degradation in  $I_{max}$  (traps?)
- Rapid degradation in diode current

- > Completion of current GCS tests: Nov. 25, 2016
- > Completion of current Lowell tests: Dec. 2, 2016
  - Preliminary results reported by Nov. 14, 2016
- > WIN process reliability: May 2017

## **Other reliability tests:**

- > Humidity acceleration model (HAST, THB)
  - Need to test ST material – is it fundamentally more robust than WIN/GCS/MTS?
  - Need to decide which process should be used for acceleration model testing